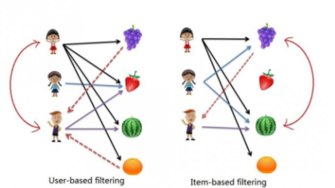
⊗ RECOMMENDER SYSTEMS — 3


User-based filtering
Item-based filtering

Agenda →

Collaborative

item item

user user

Recap

Content Based, Rec sys

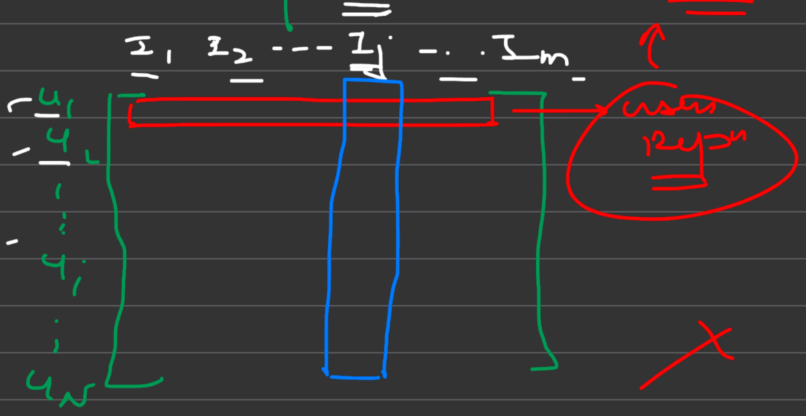
Regression Problem

code

Collaborative filtering

Vectorization

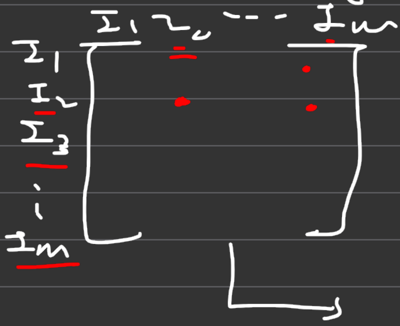
A = interaction matrix


I_1
I_2
I_j
I_m
u_1
u_2
u_i
u_N
user Repn

$$u_i = \begin{array}{|c|c|c|c|c|c|}\hline \; & \; & \; & \; & \; & \; \\ \hline \end{array} \quad (m,)$$

$$I_j = \begin{array}{|c|c|c|c|c|c|}\hline 1 & 0 & 1 & 0 & 1 & 1 \\ \hline \end{array} \quad (n,1)$$

(Similarity Item Item)



× Realtime ×

↓

updated

lookup table

(User user similarity)

item item →

User user → Which users Resemble me

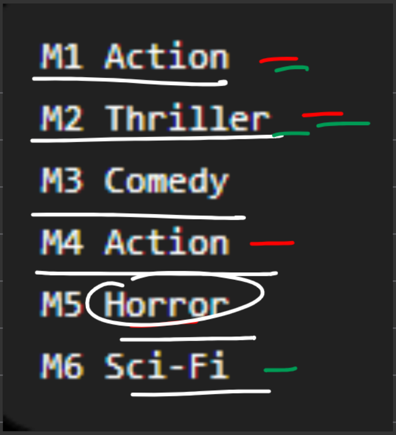
✓ A → HORROR = Conjuring

✓ B → ACTION

✓ C → suspense | thriller] xo

you like → (ACTION + suspense)

user user



M1 Action

M2 Thriller

M3 Comedy

M4 Action

M5 Horror

M6 Sci-Fi

user (k) = [1 1 0 1 0 1]

A = [0 0 0 0 1 0]

B = [1 1 0 1 0 0]

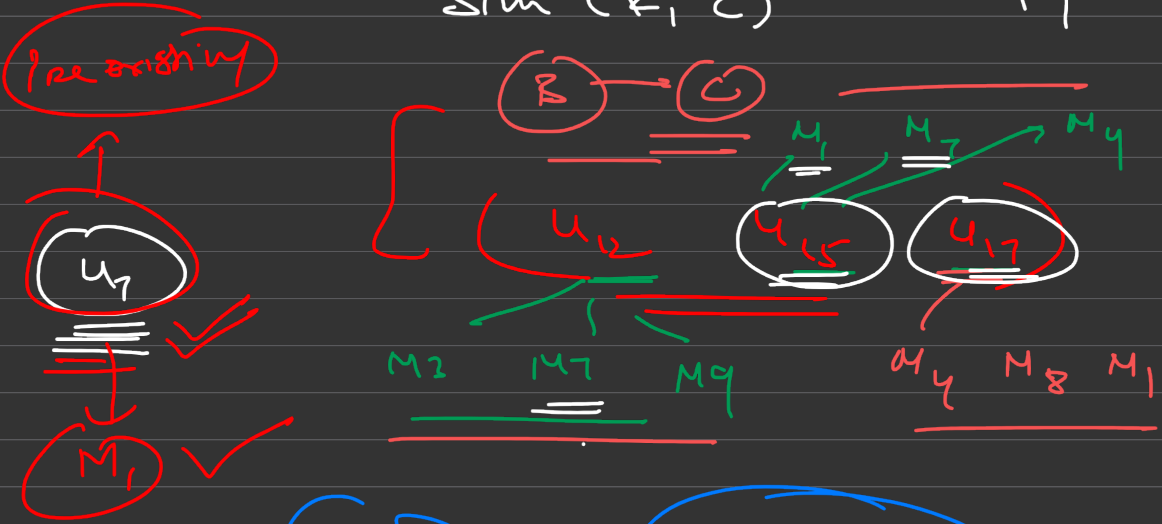
C = [0 1 0 0 0 1]

User User

sim(k, B) = 0.87

sim(k, A) = 0 ⇒

sim(k, C) = 0.71



Pre existing
U_7
M_1
B
C
M_1
M_7
M_4
U_{12}
U_{15}
U_{17}
M_3
M_7
M_9
M_4
M_8
M_1

$M_2 \to$ Recommend

New user → Netflix

M_1 M_2 M_3

$$u_i = \begin{array}{|c|c|c|c|c|} \hline 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

M movies

×

No interactions

⊛ how to find similar users?

Collaborative ⟹ Cold start

New movie ⟹ ×

$$I_j = \begin{array}{|c|c|c|c|c|c|c|} \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

⟹ Cold start problem

Collaborative

Content Based

Hybrid

dont take interactions

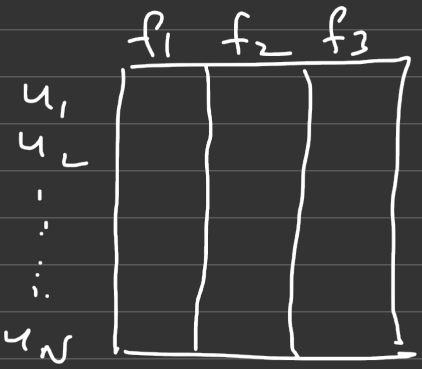
Ratings

liked / not

content → meta data

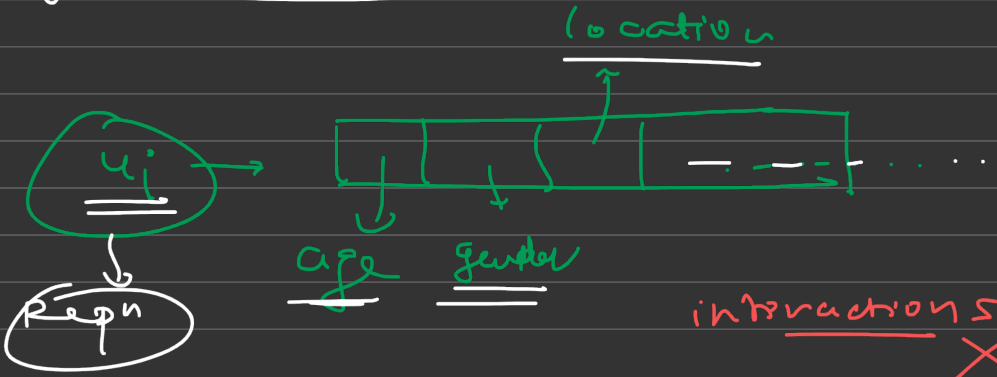
=

user



f1
f2
f3
u1
u2
⋮
uN

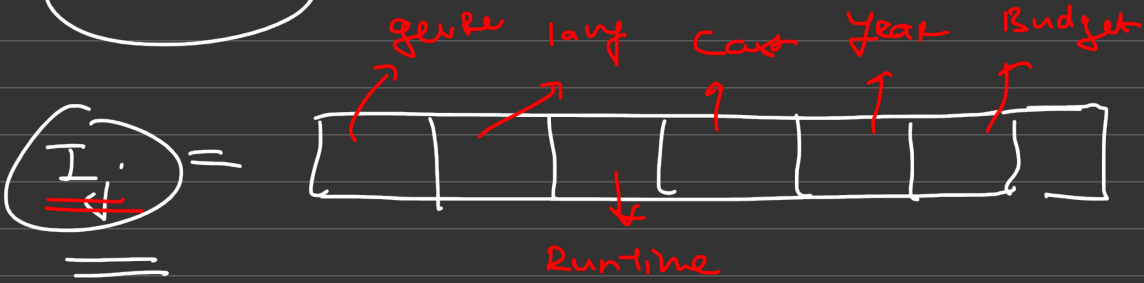
[age, gender, location, interests, language, avg watch time]



location
ui
age
gender
Repn

interactions

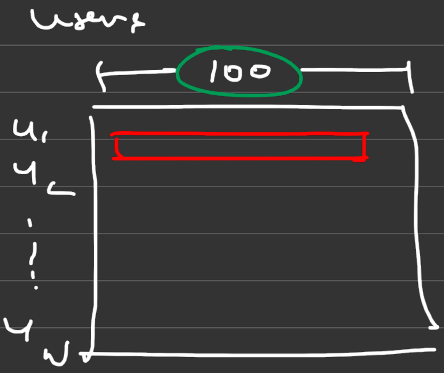
Items → movies

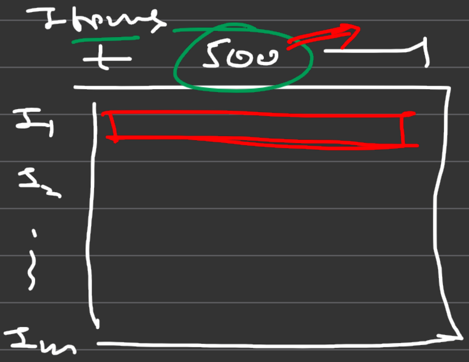

I₁
=
genre
lang
cast
year
Budget
Runtime

⟹ meta data

New movie → ~~interaction~~ ✗

→ content / meta data ✓


users
100
u₁
u₂
u_N


Items
500
I₁
I₂
I_m

user – user

item – item

CONTENT (COLLABORATIVE)

Time consuming

interactions ✓

[ ] ↓

[ ] (Create features)

(domain knowledge)

[ ]

[ ]

Pros

Cold Start ✓

New users

Cons

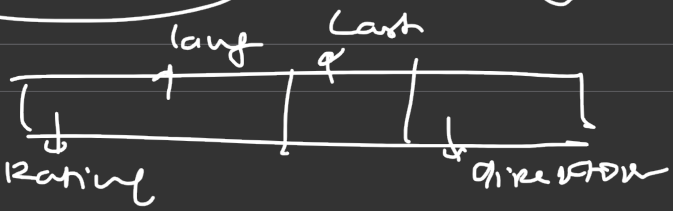
Time consuming

vector = [0 | 0 | 0 | 0]

CONTENT

New movies ✓

x ⇒


lang
Cast
Rating
director

Creating features

COLAB

CONTENT

Hybrid

User → Predefined

Hybrid

CONTENT

User → Interactions →

(Collaborative filtering)

HYBRID

Netflix → launched a new movie

Day 1 → X

collab → weak ✓

↓

1000s

1 week

Iq ✓

→

1	0	1	1	0	1

42

Collaborative

New item

↓

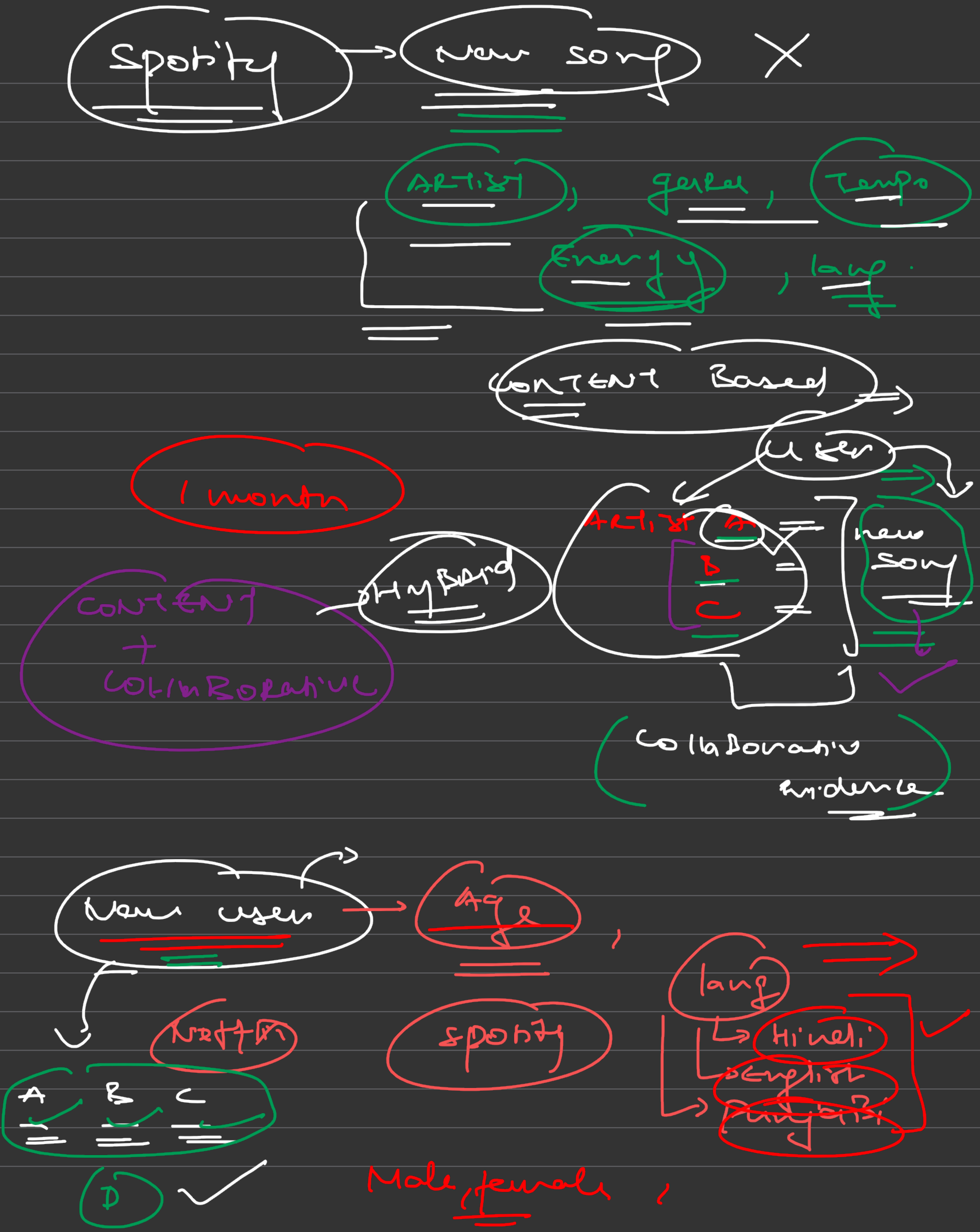
use CONTENT info

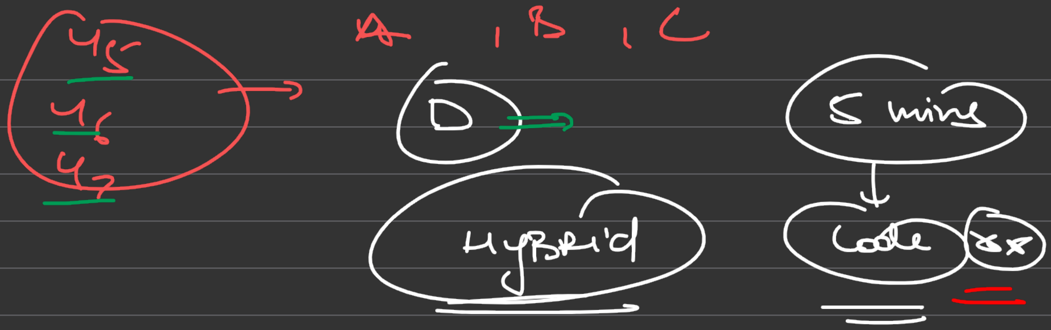
↓

Collect interactions

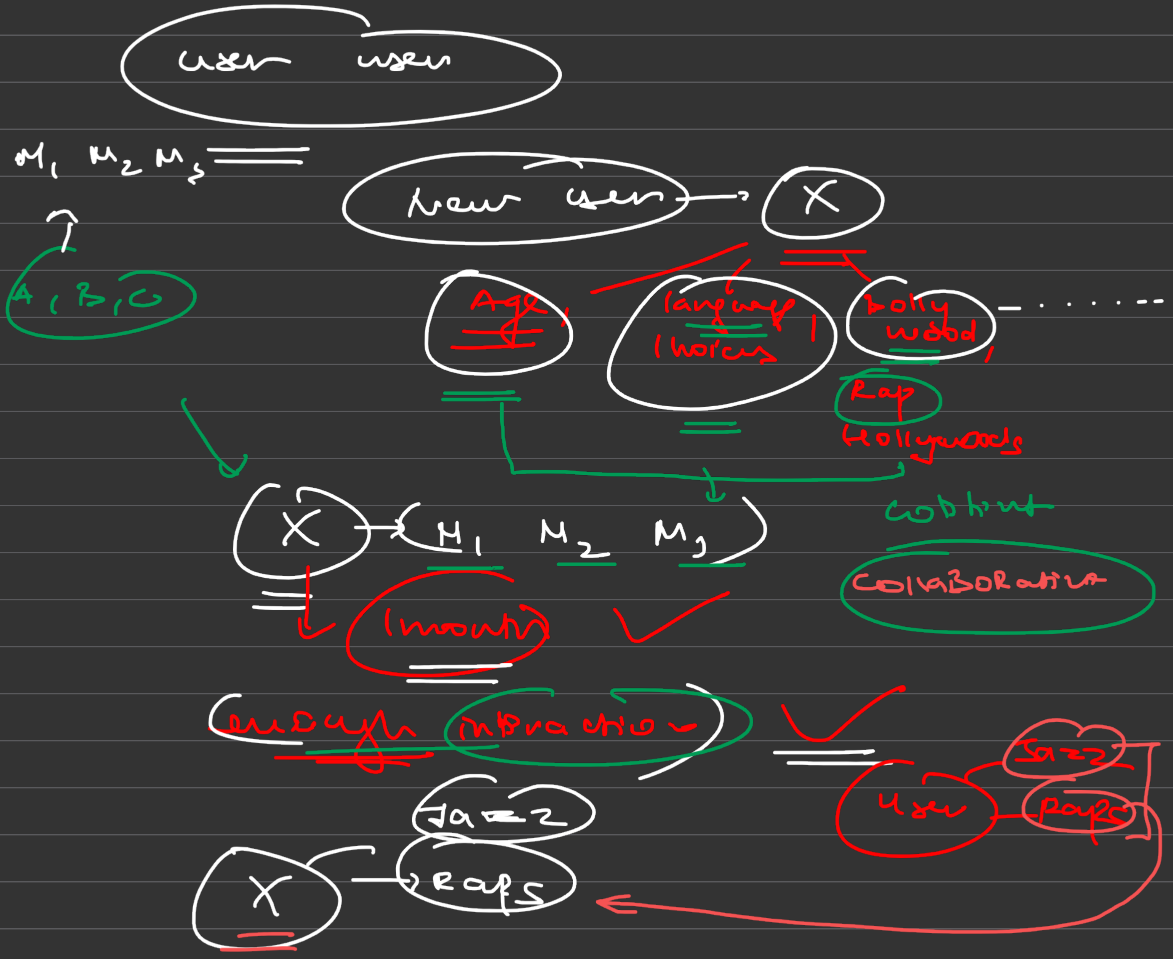
→ Collaborative

→ combine Both

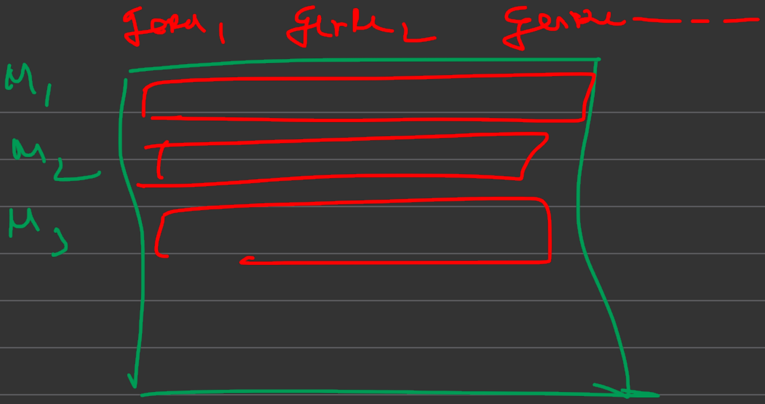

Spotify
New song
✗
ARTIST
genre
Tempo
Energy
lang
CONTENT Based
user
ARTIST
A
B
C
new song
✓
collaborative
evidence
1 month
CONTENT
+
COLLABORATIVE
Hybrid
New user
Age
spotify
lang
Hindi
English
Punjabi
Netflix
A
B
C
D
✓
Male, female

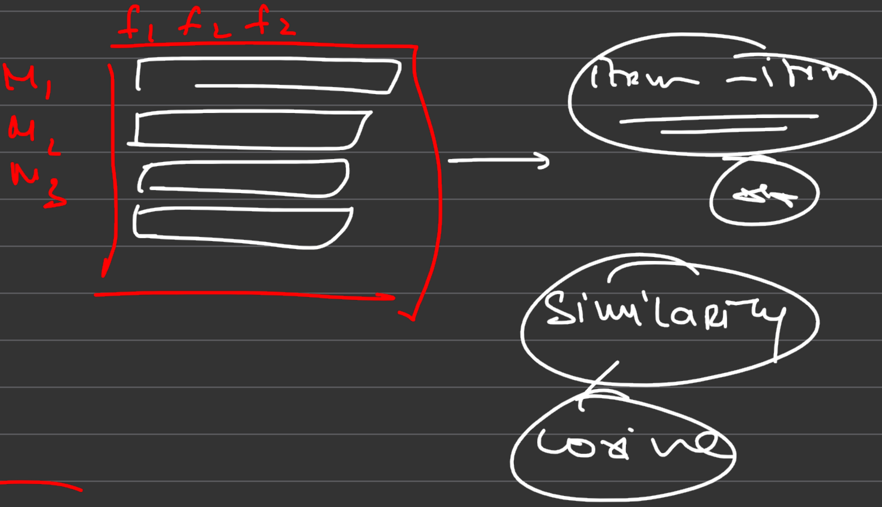


45
46
47
A, B, C
D
5 mins
Code
xx
Hybrid



user user
M1, M2, M3
A, B, C
new user
X
Age,
language / choices
bolly wood
Rap
Hollywoods
content
X
(M1 M2 M3)
1 month
enough interaction
collaborative
Jazz
X
Raps
user
Jazz
Raps

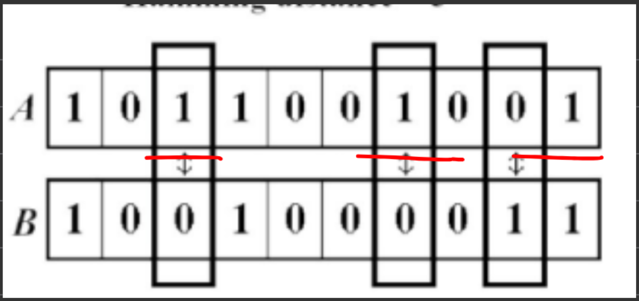

Genre1 Genre2 Genre----
M1
M2
M3


f1 f2 f2
M1
M2
M3
item - item
Similarity
cosine

Hamming distance ⟹

A

B


A 1 0 1 1 0 0 1 0 0 1
B 1 0 0 1 0 0 0 0 1 1

1 1 1 = 3

Hamming distance

Distance ↑

distance ↓ ↓

0 0 0 0

0 0 0 0

0

0 1 0 1

1 0 1 0

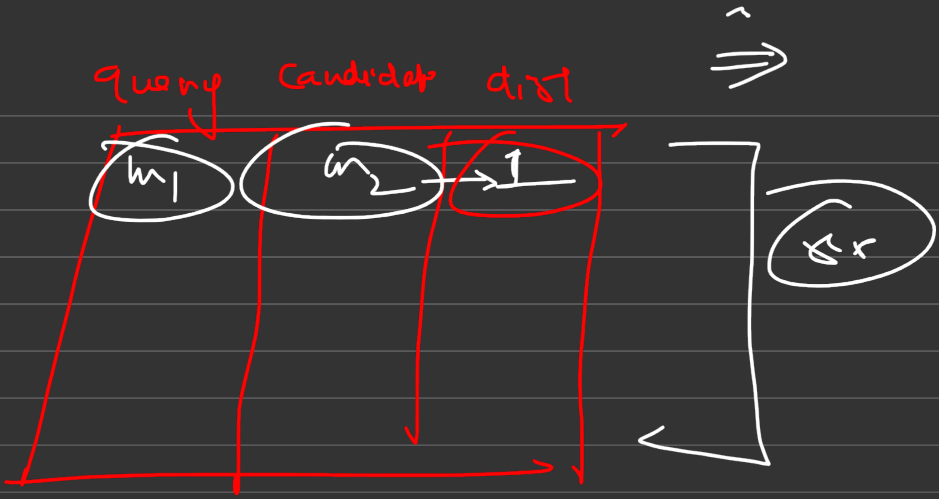
d=4

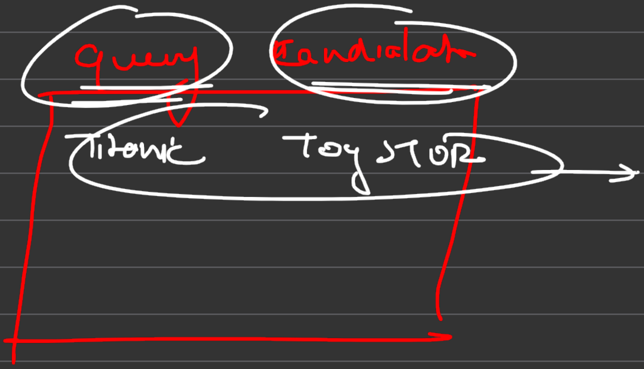
Hamming distance

Cosine → 1 → Similar

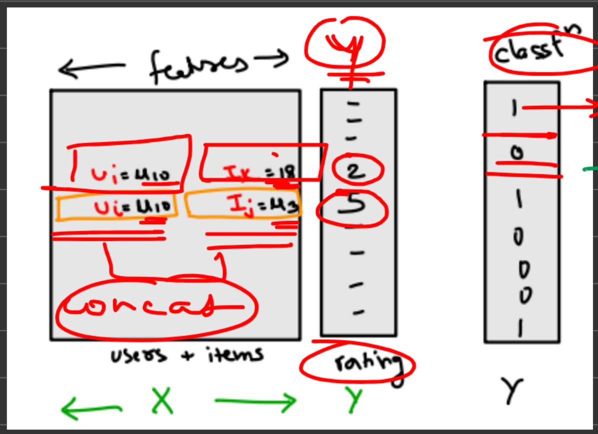
Hamming → ↓D

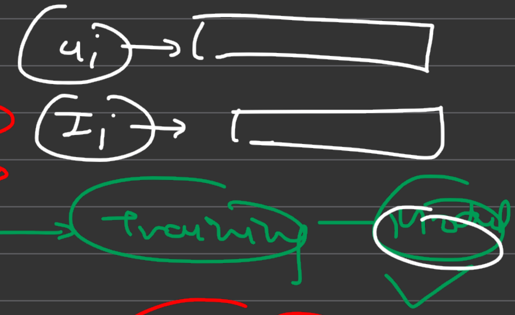
✓


query
candidate
dist
m1
m2
1
Ex


query
candidate
Titanic
Toy STOR

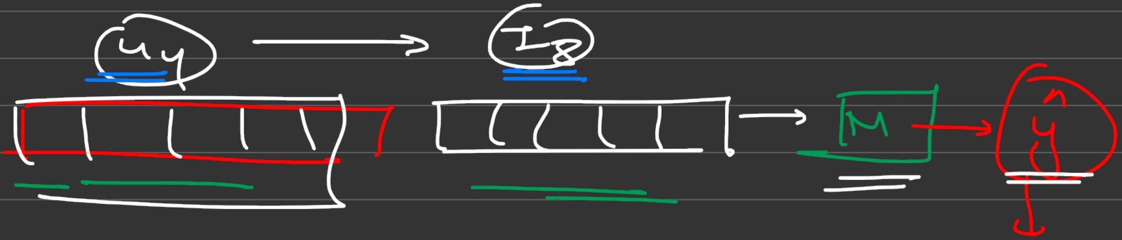
Rec sys as Regression | Classification


← features →
y
class
n
$u_i = u_{10}$
$I_k = 18$
$u_i = u_{10}$
$I_j = u_3$
2
5
concat
users + items
rating
1
0
1
0
0
0
1
← X →
Y
Y


u_i
I_j
training
model

Supervised

concat

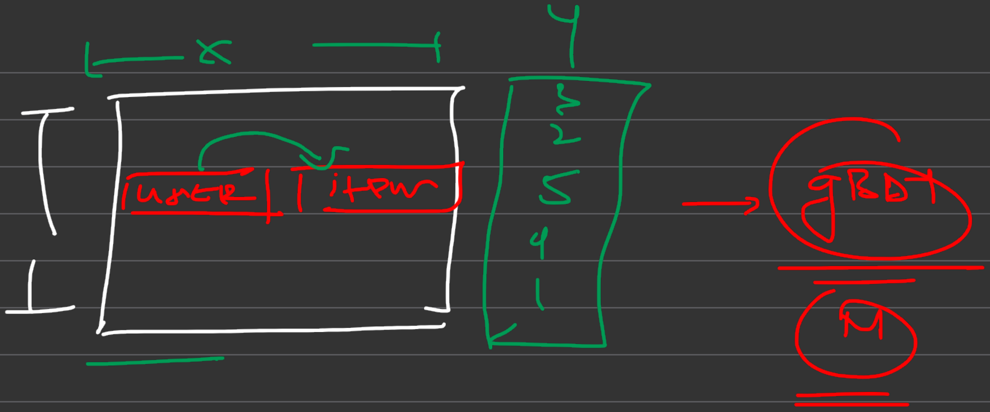


Predicted Rating

Classification

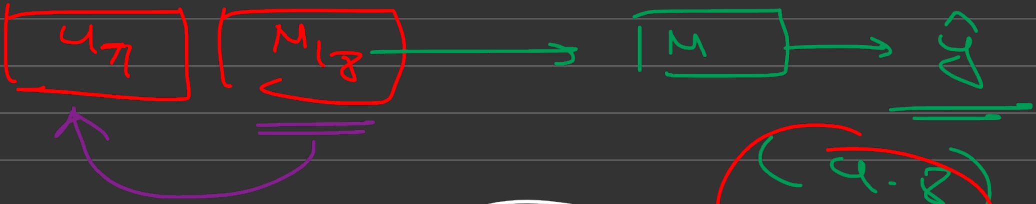
[1.6]

> Threshold


x
y
user
item
3
2
5
4
1
gBDT
M

u9

Regression



(4.8)

Doubts

Matrix Factorization